



# Quantum cutting and near-infrared emissions in $\text{Ho}^{3+}/\text{Yb}^{3+}$ codoped transparent glass-ceramics

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## ABSTRACT

Visible to near-infrared quantum cutting and 2.0  $\mu\text{m}$  near-infrared emissions have been obtained in  $\text{Ho}^{3+}/\text{Yb}^{3+}$  co-doped oxyfluoride transparent glass-ceramic, containing cubic  $\text{CaF}_2$  nanocrystals, under 473 nm excitation. The formation of  $\text{CaF}_2$  nanocrystals and the incorporation of  $\text{Ho}^{3+}/\text{Yb}^{3+}$  ions into these nanocrystals have been confirmed from x-ray diffraction and photoluminescence studies, respectively. Lifetime of 1.2  $\mu\text{m}$  emission has been enhanced by three orders of magnitude in the glass-ceramic compared to that of the precursor glass. Time-resolved near-infrared quantum cutting emission of  $\text{Yb}^{3+}$  ions at different delay times under 532 nm laser excitation has also been obtained. Studies reveal that the infrared quantum cutting emission of  $\text{Yb}^{3+}$  and  $\text{Ho}^{3+}$  ions in the  $\text{CaF}_2$  nanocrystalline glass-ceramic could improve the efficiency of crystalline-Si solar cells, and that the 1.2  $\mu\text{m}$  emission could be useful for further enhancing transmission windows using optical amplifiers.

## 1. Introduction

A transparent oxyfluoride glass-ceramic (TGC) is a two-phase material containing fluoride nanocrystals that precipitate in an amorphous environment during the controlled thermal treatment of the precursor glass. In recent years, TGCs have attracted great deal of interest due to their potential applications in diverse fields of science and technology [1–3]. Alumina-silicate based glasses are most suitable for precipitation of fluoride nanocrystals due to their excellent thermal, chemical and mechanical properties compared to phosphate or fluoride glasses. Further, in these TGCs, the host for the trivalent lanthanide ( $\text{Ln}^{3+}$ ) ions corresponds to fluoride crystals with low phonon energy and, hence, low multiphonon probabilities, which results in increasing the luminescence efficiency of these materials compared to their precursor glasses [3].  $\text{Ln}^{3+}$  ions doped TGCs are widely investigated as they exhibit both upconversion (UC) and downconversion (DC) processes besides excellent  $\text{Ln}^{3+}$  ion solubility [4]. Recently, TGCs containing fluoride nanocrystals have been explored for laser cooling applications as they show

extraordinary properties, including high photoluminescence quantum yield and low background absorption [5–7].

Trivalent holmium,  $\text{Ho}^{3+}$  ion is suitable for infrared-to-visible upconversion since it has two pumping energy levels,  $^5\text{F}_5$  and  $^5\text{I}_4$ , along with a relatively long lived intermediate energy level,  $^5\text{I}_7$ , that acts as a good population reservoir for upconversion processes. As a result, in recent times there is significant amount of interest in  $\text{Ho}^{3+}$ -doped glasses and crystals as possible upconversion materials. Further,  $\text{Ho}^{3+}$ -doped glass fibers are found to emit continuous wave upconversion laser under red light pumping [8]. In addition,  $\text{Yb}^{3+}$  ion has a relatively simple energy level structure with the  $^2\text{F}_{7/2}$  level as ground state and the  $^2\text{F}_{5/2}$  level as excited state. Its broad emission, ranging between 900 and 1100 nm, overlaps with the crystalline silicon's band gap of  $\sim 1.12$  eV, making  $\text{Yb}^{3+}$  ion an important candidate for near-infrared (NIR) quantum cutting (QC) materials to improve the efficiency of crystalline-silicon (c-Si) solar cells [9]. Materials co-doped with  $\text{Yb}^{3+}$  ions and other  $\text{Ln}^{3+}$  ions are attractive for the development of upconversion devices. In the specific case of  $\text{Ln}^{3+}$  ( $\text{Er}^{3+}$ ,  $\text{Tm}^{3+}$  or  $\text{Ho}^{3+}$ ) ions, resonance energy transfer is

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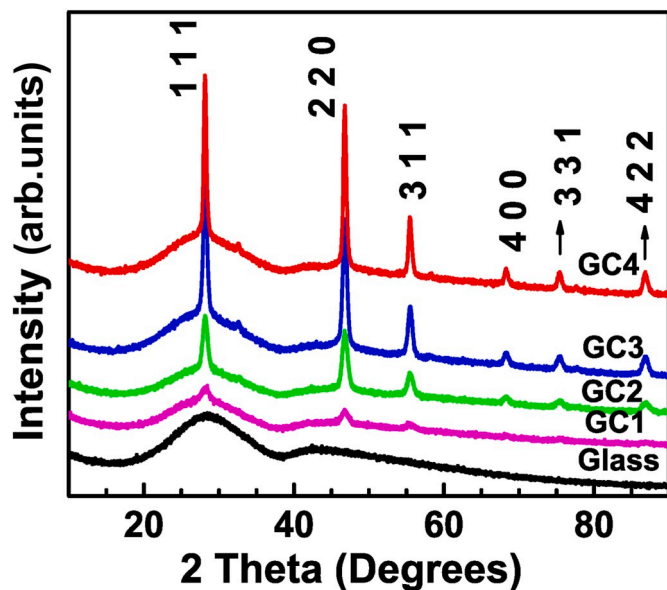


Fig. 1. X-ray diffraction patterns of the oxyfluoride glass and glass-ceramics.

facilitated by the energy matching of their excited states with that of  $\text{Yb}^{3+}$  ion [10]. The  $\text{Yb}^{3+}$  ion is found to be efficient sensitizer for  $\text{Ho}^{3+}$  lasers [2], due to considerable energy match between  $\text{Yb}^{3+}$  and  $\text{Ho}^{3+}$  ions and hence,  $\text{Ho}^{3+}/\text{Yb}^{3+}$  couple is a good choice for NIR QC investigations [11]. The NIR QC, which converts one UV-blue photon into two NIR photons of  $\sim 1.0 \mu\text{m}$ , has attracted more attention for their application in plasma display, mercury free fluorescent tubes, and solar cells [11,12]. In recent times, more emphasis has been focused on NIR QC in different systems with the combination of different  $\text{Ln}^{3+}$  ions with  $\text{Yb}^{3+}$  ions, for example,  $\text{Nd}^{3+}/\text{Yb}^{3+}$  [13],  $\text{Tb}^{3+}/\text{Yb}^{3+}$  [14–17],  $\text{Er}^{3+}/\text{Yb}^{3+}$  [18,19],  $\text{Tm}^{3+}/\text{Yb}^{3+}$  [20],  $\text{Pr}^{3+}/\text{Yb}^{3+}$  [13,21]  $\text{Ho}^{3+}/\text{Yb}^{3+}$  [9, 22,23], and  $\text{Yb}^{2+}/\text{Yb}^{3+}$  [24] to enhance the efficiency of solar cells. On the other side, solid state laser working in the infrared region at  $2 \mu\text{m}$  is attracting increasing interest for its wide range of applications such as remote sensing, optical parametric oscillators, eye safe LIDAR, biomedical applications, environmental sensing, etc. [2,25]. Among the  $\text{Ln}^{3+}$  ions,  $\text{Ho}^{3+}$  is one of the important active ions that can emit at  $2 \mu\text{m}$  through the  $^5\text{I}_7 \rightarrow ^5\text{I}_8$  transition.

In the present work,  $\text{Ho}^{3+}/\text{Yb}^{3+}$  co-doped TGCs containing  $\text{CaF}_2$  nanocrystals have been synthesized through post-thermal treatment method and their visible, QC and NIR emissions have been measured exciting at 473 nm. NIR QC emissions are observed at 1.0 and  $1.2 \mu\text{m}$  wavelengths under 473 nm excitation and their possible mechanisms are discussed. The NIR emission spectra at  $2.0 \mu\text{m}$  have been measured in glass and GCs and explained their variation with respect to size of the nanocrystals. Absorption and emission cross-sections and figure of merit for the  $1.2 \mu\text{m}$  and  $2.0 \mu\text{m}$  emissions have also been calculated.

## 2. Experimental details

Precursor  $\text{Ho}^{3+}/\text{Yb}^{3+}$  co-doped oxyfluoride glass, with composition (in mol%)  $45\text{SiO}_2\text{--}20\text{Al}_2\text{O}_3\text{--}8\text{CaO--}24\text{CaF}_2\text{--}2\text{Yb}_2\text{O}_3\text{--}1\text{Ho}_2\text{O}_3$ , was prepared by the melt-quenching technique at  $1400^\circ\text{C}$  for 2 h using a covered platinum crucible. The melt was poured onto a pre-heated brass mold kept just below the glass transition temperature ( $T_g$ ), at which glass samples were annealed, and slowly cooled to room temperature to avoid thermal stress and strains. Finally, glass samples were annealed for 4h at 650, 675, 700 and  $725^\circ\text{C}$  (hereafter referred to as GC1, GC2, GC3 and GC4, respectively), close to the onset of crystallization temperature  $T_c$  ( $675^\circ\text{C}$ ) of  $\text{CaF}_2$  nanocrystals to transform them into glass-ceramics. Structural differences between the precursor glass and glass-ceramics were identified by using X-ray diffraction technique, performed with a

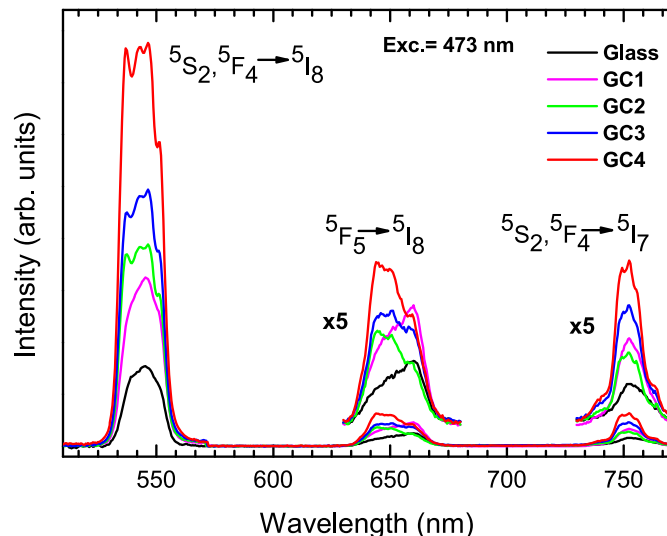


Fig. 2. Visible emissions of  $\text{Ho}^{3+}$  ions in the oxyfluoride glass and glass-ceramics (GC) under 473 nm laser excitation.

Thermo ARL X'TRA powder diffractometer with  $\text{Cu-K}\alpha$  radiation ( $1.54 \text{ \AA}$ ).

Visible and NIR quantum cutting emissions were obtained by pumping with a cw laser at 473 nm and the emission was collimated and focalized in an optical fiber coupled to a ANDOR spectrograph, a SR-303i-B equipped with a Newton 970EMCCD camera for the visible range, and an SR-500i-B2 with a iDus InGaAs photodiode array for the NIR range. The time-resolved fluorescence was obtained by exciting the samples with an EKSPLA 10 ns pulsed Optical Parametric Oscillator (OPO) laser and recording the luminescence with a Jobin-Yvon 320 spectrometer equipped with a Hamamatsu NIR-extended photomultiplier connected to a Lecroy digital storage oscilloscope.

## 3. Results and discussion

### 3.1. X-ray diffraction spectra

X-ray diffraction (XRD) patterns of  $\text{Ho}^{3+}/\text{Yb}^{3+}$ -doped precursor glass and glass-ceramics are shown in Fig. 1. The profile of the precursor glass contains two broad curves typical of an amorphous structure, whereas the GCs contain several additional sharp peaks related to the diffraction pattern of a crystalline structure, which can be identified as the  $\text{CaF}_2$  cubic phase with  $Pm\bar{3}m$  ( $\alpha = \beta = \gamma = 90^\circ$ ) space group, by comparing with standard data (JCPDS Card No. 35–0816). These crystals form nanostructures that are found to precipitate from the precursor aluminosilicate glass through a suitable thermal treatment that, from the XRD peak widths and using the Scherer's formula [26], sizes of the nanocrystals are found to be around 8, 10, 13 and 18 nm for GC1, GC2, GC3 and GC4, respectively. Samples exhibit only the cubic phase and no second phase is detected in the XRD pattern, indicating the successful formation of  $\text{CaF}_2$  nanocrystals.

### 3.2. Emission spectra and decay curves

#### 3.2.1. Visible emission spectra

Visible emission spectra of glass and glass-ceramics (GCs) are measured exciting the  $^5\text{I}_8 \rightarrow ^5\text{F}_3$  transition of the  $\text{Ho}^{3+}$  ion under 473 nm laser excitation, and are shown in Fig. 2. Spectra consist of three bands centered at  $\sim 542$ ,  $\sim 648$  and  $\sim 752 \text{ nm}$  corresponding to the  $^5\text{S}_2, ^5\text{F}_4 \rightarrow ^5\text{I}_8$ ,  $^5\text{F}_5 \rightarrow ^5\text{I}_8$  and  $^5\text{S}_2, ^5\text{F}_4 \rightarrow ^5\text{I}_7$  transitions, respectively. When samples are excited,  $\text{Ho}^{3+}$  ions reach the  $^5\text{F}_3$  level, from which a fast non-radiative decay takes place and ions reach the  $^5\text{S}_2, ^5\text{F}_4$  metastable multiplets, resulting in green emission at  $\sim 542 \text{ nm}$  and red emission at

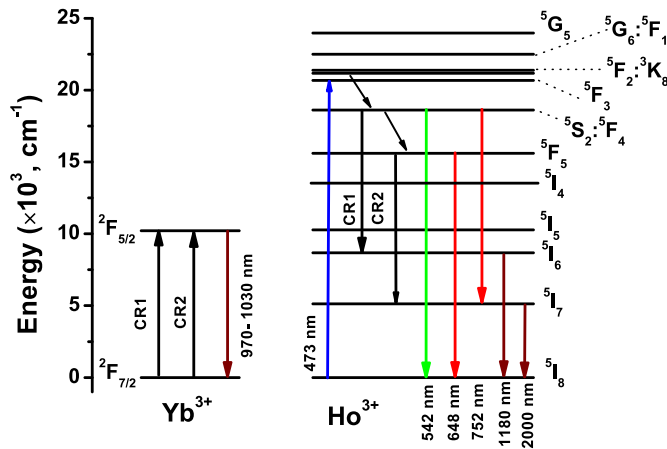


Fig. 3. Partial energy level diagram of the  $\text{Ho}^{3+}/\text{Yb}^{3+}$  co-doped oxyfluoride glass and glass-ceramics. CR indicates a cross-relaxation channel.

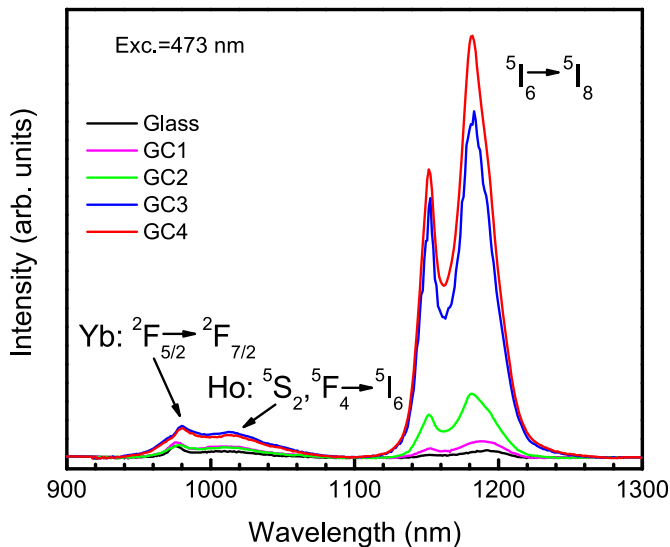
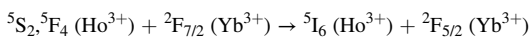


Fig. 4. Quantum cutting emissions of  $\text{Yb}^{3+}$  and  $\text{Ho}^{3+}$  ions in the oxyfluoride Glass and glass-ceramics (GC) under 473 nm laser excitation.

~752 nm. From the  $^5\text{S}_2, ^5\text{F}_4$  levels further non-radiative deexcitation processes populates the  $^5\text{F}_5$  level, resulting in  $^5\text{F}_5 \rightarrow ^5\text{I}_8$  emission at ~648 nm. Mechanisms of excitation and emissions are shown in Fig. 3. Lin et al. [27] obtained similar emission spectra in  $\text{Ho}^{3+}/\text{Yb}^{3+}$  co-doped GC with  $\text{YF}_3$  nanocrystals under 488 nm excitation. Out of these three transitions, the green emission is more intense than the other two. As can be seen from Fig. 2, the emission intensity and splitting of the bands increases from glass to GC4, indicating the progressive incorporation of  $\text{Ho}^{3+}$  ions into the  $\text{CaF}_2$  nanocrystals and the increase in size of the nanocrystals.

### 3.2.2. Quantum cutting and 1.2 $\mu\text{m}$ emission spectra

As already mentioned, when the samples are excited with a 473 nm cw laser, the  $\text{Ho}^{3+}$  ions are excited to the  $^5\text{F}_3$  level, followed by a non-radiative relaxation to the  $^5\text{S}_2, ^5\text{F}_4$  level. From this level, the following resonant cross-relaxation energy transfer, denoted as CR1 in Fig. 3, occurs:



and one  $\text{Ho}^{3+}$  1180 nm photon (from the  $^5\text{I}_6$  level) and one  $\text{Yb}^{3+}$  980 nm photon (from the  $^2\text{F}_{5/2}$  level) are emitted.

Further, non-radiative multiphonon relaxations from the  $^5\text{S}_2, ^5\text{F}_4$

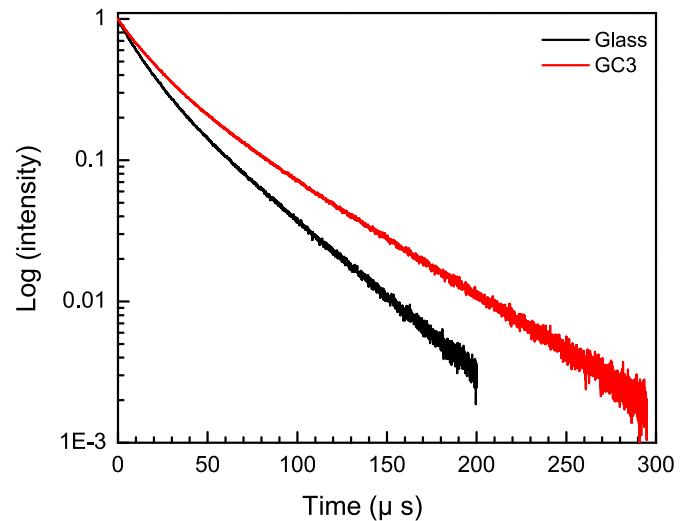
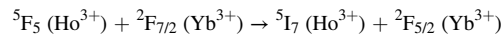


Fig. 5. Decay curves of the  $\text{Yb}^{3+}$  ions in the oxyfluoride glass and glass-ceramic (GC3) exciting at 944 nm and monitoring the emission at 1020 nm, where there is shoulder in the  $\text{Yb}^{3+}$  emission.

levels populate the  $^5\text{F}_5$  one and, from this level, another resonant cross-relaxation energy transfer, shown as CR2 in Fig. 3, takes place as follows:



resulting in the emission of one  $\text{Ho}^{3+}$  2.0  $\mu\text{m}$  photon (from the  $^5\text{I}_7$  level) and one  $\text{Yb}^{3+}$  980 nm photon (from the  $^2\text{F}_{5/2}$  level). All these excitations, CR and emissions channels are shown in the partial energy level diagram of  $\text{Yb}^{3+}$  and  $\text{Ho}^{3+}$  ions in Fig. 3. It is worth noting that Zhou et al. [26] and Lin et al. [27] proposed similar mechanisms to explain QC processes in  $\text{Ho}^{3+}/\text{Yb}^{3+}$  codoped glass-ceramics.

Emissions from  $\text{Yb}^{3+}$  ( $^2\text{F}_{5/2} \rightarrow ^2\text{F}_{7/2}$ ) and  $\text{Ho}^{3+}$  ( $^5\text{S}_2, ^5\text{F}_4 \rightarrow ^5\text{I}_6$ ) ions overlap giving a wide band in the range of 950–1100 nm, as shown in Fig. 4. Additionally,  $\text{Yb}^{3+}$  emissions consist of two peaks, a sharp one at ~980 nm and broader one at ~1020 nm, attributed to emission from the lowest Stark level of  $^2\text{F}_{5/2}$  multiplet to the lowest and next higher Stark levels of the  $^2\text{F}_{7/2}$  ground multiplet, respectively [28]. The intensity of  $\text{Yb}^{3+}$  emissions increases, due to the change of the  $\text{Ln}^{3+}$  local structure, from glass to glass-ceramic GC3, and then decreases marginally for GC4, which could be due to back transfer processes from  $\text{Yb}^{3+}$  to  $\text{Ho}^{3+}$  ions. The  $^5\text{I}_6 \rightarrow ^5\text{I}_8$  transition gives emission at ~1.2  $\mu\text{m}$  and is also shown in Fig. 4 for the glass and GCs. Their intensities are higher compared to those of 1.0  $\mu\text{m}$ , and increase monotonically from the glass to the glass-ceramic GC4 with the increase in size of the nanocrystals and the progressive incorporation of  $\text{Ho}^{3+}$  ions into low phonon energy  $\text{CaF}_2$  nanocrystals.

As can be seen from Figs. 2 and 4, in moving from glass to GC4, the enhancement factor of the intensity is higher for the  $^5\text{I}_6 \rightarrow ^5\text{I}_8$  emission band compared to the  $^5\text{S}_2, ^5\text{F}_4 \rightarrow ^5\text{I}_7$  band of  $\text{Ho}^{3+}$  ions. This can be explained taking into account that the energy difference between the  $^5\text{S}_2, ^5\text{F}_4$  and  $^5\text{F}_5$  levels is ~3020  $\text{cm}^{-1}$  whereas the difference between the  $^5\text{I}_6$  and  $^5\text{I}_7$  levels is ~3540  $\text{cm}^{-1}$  (from absorption spectrum). Larger energy gap results in lower multiphonon relaxation from the upper level. In moving from glass to GC4, phonon energy of the host progressively decreases. As a result of this, there is progressive increase in radiative emission from the  $^5\text{I}_6$  level compared to the  $^5\text{S}_2, ^5\text{F}_4$  level in moving from glass to GC4. Another reason for higher enhancement factor of the  $^5\text{I}_6 \rightarrow ^5\text{I}_8$  emission could be increase in CR1 process, in moving from glass to GC4, which depopulates the  $^5\text{S}_2, ^5\text{F}_4$  level and populates the  $^5\text{I}_6$  level.

Quantum cutting (QC) is a process in which one photon of high energy converts into two photons of lower energy. This process in the  $\text{Ho}^{3+}$  ions, where each photon at 473 nm that is absorbed can produce two NIR photons, given by  $^2\text{F}_{5/2} \rightarrow ^2\text{F}_{7/2}$  &  $^5\text{I}_6 \rightarrow ^5\text{I}_8$  (1.0 & 1.2  $\mu\text{m}$ ) and  $^2\text{F}_{5/2}$

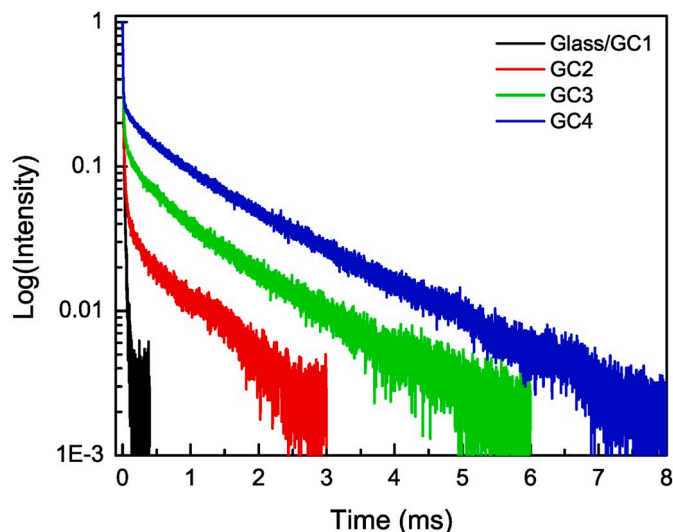


Fig. 6. Decay curves of the  $\text{Ho}^{3+}$  ion in the oxyfluoride glass and glass-ceramics (GC) exciting at 473 nm and monitoring the emission at 1.2  $\mu\text{m}$ .

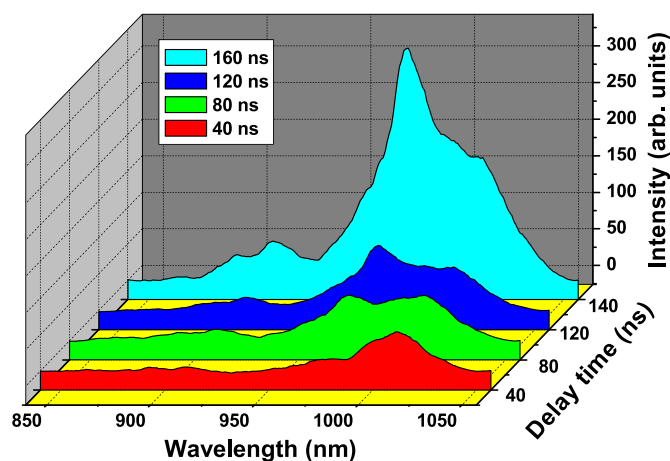


Fig. 7. Time-resolved near-infrared quantum cutting emissions of  $\text{Yb}^{3+}$  ions at different delay times after a 532 nm pulsed-laser excitation in a glass-ceramic (GC3).

$\rightarrow {}^2\text{F}_{7/2}$  &  ${}^5\text{I}_7 \rightarrow {}^5\text{I}_8$  (1.0 & 2.0  $\mu\text{m}$ ). The low energy photons ( $\sim 1.2$  and  $\sim 2.0$   $\mu\text{m}$ ) are more intense and efficient as they occur only between  $\text{Ho}^{3+}$  ions, whereas the former ones ( $\sim 1.0$   $\mu\text{m}$ ) involves energy transfer from  $\text{Ho}^{3+}$  to  $\text{Yb}^{3+}$  ions and, hence, they are weaker.

The effect of  $\text{Yb}^{3+}$  ions can be analyzed using the decay curves of the  ${}^2\text{F}_{5/2}$  level of  $\text{Yb}^{3+}$  ions in glass and GC3. These decays are measured by exciting them with a 944 nm laser radiation and monitoring the emission at  $\sim 1020$  nm, where there is shoulder in the  $\text{Yb}^{3+}$  ions emission, and are shown in Fig. 5. Both decay curves are clearly non-exponential and the average lifetimes [11] are found to be  $\sim 15$  and  $\sim 28$   $\mu\text{s}$  for glass and GC3, respectively. It is interesting to note that the lifetime in GC3 increases significantly, almost by a factor of 2, due to the presence of  $\text{Yb}^{3+}$  ions in a low phonon energy  $\text{CaF}_2$  nanocrystalline environment.

Decay curves for the  ${}^5\text{I}_6 \rightarrow {}^5\text{I}_8$  transition of  $\text{Ho}^{3+}$  ions at 1.2  $\mu\text{m}$  measured in the glass and the glass-ceramics, under 473 nm laser excitation, are shown in Fig. 6. As can be seen, all decay curves are non-exponential in nature and consists of short and long components. The short component corresponds to ions that reside in the glass matrix and are not incorporated in the nanocrystals. But, as can be seen from Fig. 6, this component is reduced from GC1 to GC4. Lifetime of the long component of 1.2  $\mu\text{m}$  decay, which is due to ions present in  $\text{CaF}_2$

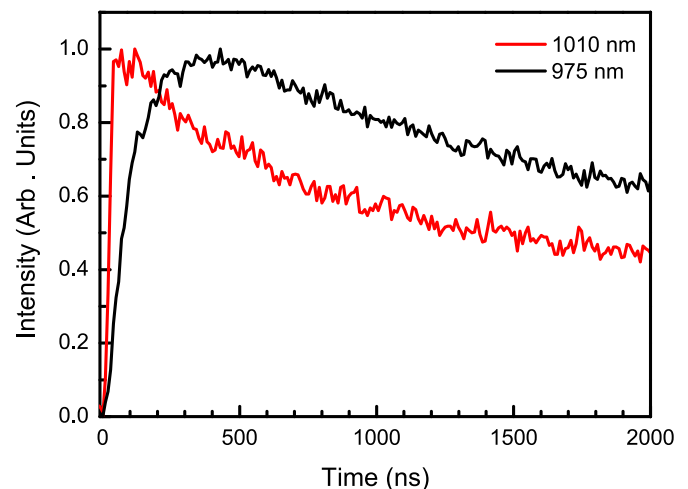


Fig. 8. Decay curves of  $\text{Yb}^{3+}$  (975 nm) and  $\text{Ho}^{3+}$  (1010 nm) ions in glass-ceramic (GC3) under 532 nm pulsed laser excitation.

nanocrystals, increases with the increase in the size of nanocrystals from 15  $\mu\text{s}$  (Glass/GC1) to 1.586 ms (GC4). It is worth noting that the lifetime in GC4 is three orders of magnitude ( $\sim 1006$  times) longer compared to that of the glass, due to the presence of  $\text{Ho}^{3+}$  ions in low phonon energy  $\text{CaF}_2$  nanocrystals. The present lifetime of the 1.2  $\mu\text{m}$  emission of  $\text{Ho}^{3+}$  ion in GC4 is longer compared to other reported  $\text{Ho}^{3+}$  doped matrices, such as  $\text{CaSc}_2\text{O}_4$  ceramic [29] or water-free fluorotellurite glasses [30].

Fig. 7 shows the time-resolved NIR QC emission of  $\text{Yb}^{3+}$  ions at different delay times after exciting with a 532 nm laser light, resonant with the  ${}^5\text{I}_8 \rightarrow {}^5\text{F}_4, {}^5\text{S}_2$  transition of  $\text{Ho}^{3+}$  ions. The emission spectra consist of overlapped bands at  $\sim 894$ ,  $\sim 911$ ,  $\sim 980$  and  $\sim 1020$  nm due to Stark's splitting of the excited  $\text{Yb}^{3+}({}^2\text{F}_{5/2})$  multiplet and the  ${}^5\text{F}_4, {}^5\text{S}_2 \rightarrow {}^5\text{I}_6$  transition of  $\text{Ho}^{3+}$  ions at  $\sim 1010$  nm. The energy transfer from  $\text{Ho}^{3+}$  ions is responsible for the emission of  $\text{Yb}^{3+}$  ions. The emission intensity and full width at half maximum of the 980 nm emission band increases with the increase in delay time, due to population differences among the thermally coupled levels of  $\text{Yb}^{3+}$  ions in the matrix.

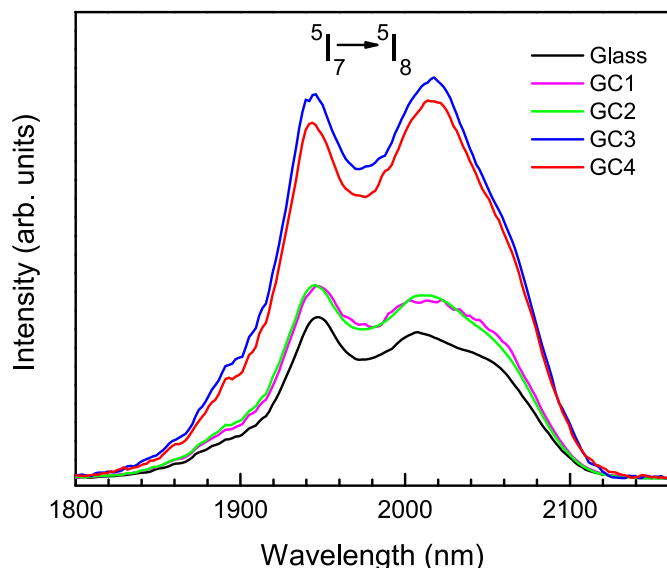
In order to study the energy transfer process more accurately, luminescence decay curves are obtained by exciting  ${}^5\text{I}_8 \rightarrow {}^5\text{F}_4, {}^5\text{S}_2$  transition at 532 nm of the  $\text{Ho}^{3+}$  ions and monitoring the emissions at  $\sim 975$  and  $\sim 1010$  nm, respectively, as shown in Fig. 8. The decay curves indicate that the energy transfer process is predominant at higher delay times because the  $\text{Yb}^{3+}$  ions are excited through CR1 or CR2 processes from  $\text{Ho}^{3+}$  ions. The rising edge in the decay curve (975 nm) of Fig. 8 indicates the slow population process of the  ${}^2\text{F}_{5/2}$  level of  $\text{Yb}^{3+}$  ions through energy transfer from  $\text{Ho}^{3+}$  ions. This provides a strong evidence for the energy transfer from the  $\text{Ho}^{3+}$  to  $\text{Yb}^{3+}$  ions [31]. On the contrary, in the emission at  $\sim 1010$  nm, associated with the  ${}^5\text{F}_4, {}^5\text{S}_2 \rightarrow {}^5\text{I}_6$  transition, predominates the emission coming from  $\text{Ho}^{3+}$  ions initially excited.

The emission cross-section between a high energy level and a lower energy level can be derived using the McCumber [32] theory,

$$\sigma_{em}(\lambda) = \sigma_{ab}(\lambda) \frac{Z_l}{Z_u} \exp\left(\frac{h\nu - E_{ZL}}{k_B T}\right) \quad (1)$$

where  $\sigma_{ab}$  is absorption cross-section,  $\sigma_{ab} = \alpha(\lambda)/N$ , where  $\alpha(\lambda)$  is the experimental absorption coefficient and  $N$  is the concentration of  $\text{Ho}^{3+}$  ions;  $Z_l$  and  $Z_u$  are the partial functions of lower and upper levels, respectively;  $k_B$  is Boltzmann's constant;  $E_j - E_i = h\nu - E_{ZL}$  is the energy separation between the two levels; and  $E_{ZL}$  is the 'zero line' energy, i.e. the energy of the transition from the lowest Stark sublevels of upper and lower manifold.

Absorption and emission cross-sections ( $\times 10^{-20}$   $\text{cm}^2$ ) are calculated for the 1.2  $\mu\text{m}$  emission of the  $\text{Ho}^{3+}$  ion and are found to be 0.93 and 5.2 for the precursor glass and 1.16 and 6.7 for the glass-ceramic GC4.



**Fig. 9.** Near infrared emissions of  $\text{Ho}^{3+}$  ions in the oxyfluoride glass and glass-ceramics (GC) under 473 nm laser excitation.

Values of emission-cross-sections of glass and GC4 are much larger than those of TZNF60 glass ( $0.63 \times 10^{-20} \text{ cm}^2$ ) [30] and oxyfluoride nano glass-ceramic ( $0.27 \times 10^{-20} \text{ cm}^2$ ) [33]. Figure of merit (FOM), defined as the product of experimental lifetime and emission cross-section [34], for the 1.2  $\mu\text{m}$  emission is found to be  $0.08 \times 10^{-23} \text{ cm}^2 \cdot \text{s}$  for glass and  $10720 \times 10^{-23} \text{ cm}^2 \cdot \text{s}$  for the glass-ceramic GC4. The FOM for GC4 is found to be about 1000 times higher than that of glass. The higher value of FOM indicates higher gain for the 1.2  $\mu\text{m}$  emission in GC4.

### 3.2.3. 2.0 $\mu\text{m}$ emission spectra

NIR emission spectra of the oxyfluoride glass and the glass-ceramics at around 2.0  $\mu\text{m}$  are measured by exciting  $^5\text{I}_8 \rightarrow ^5\text{F}_3$  transition using a 473 nm laser light and are shown in Fig. 9. As can be seen, the intensity of the bands increases from the glass to the glass-ceramic GC3 and then decreases slightly for GC4. This could be due to decrease in the magnitude of the cross-relaxation processes, CR2 in GC4, which feeds  $^5\text{I}_7$  level, due to decrease in multiphonon relaxation from  $^5\text{S}_2, ^5\text{F}_4$  to  $^5\text{F}_5$  level. The full width at half maximum (FWHM) of the all the emissions of the present study, from glass to GC4, is found to be around 153 nm ( $386 \text{ cm}^{-1}$ ), which is slightly larger than 149 nm found in TZNF60 glass [26]

but less than 171 nm found in germinate glass [35].

Absorption and emission cross-sections ( $\times 10^{-20} \text{ cm}^2$ ) of the 2.0  $\mu\text{m}$  band calculated using Eq. (1) are found to be 2.1 and 5.39 for the precursor glass and 2.09 and 5.79 for the glass-ceramic GC4, respectively. The absorption and emission cross-sections ( $\times 10^{-20} \text{ cm}^2$ ) of the present study are larger than those found in oxyfluoride glass ( $\sigma_{\text{ab}} = 0.498$ ,  $\sigma_{\text{em}} = 0.47$ ) and glass-ceramic GC C ( $\sigma_{\text{ab}} = 0.56$ ,  $\sigma_{\text{em}} = 0.66$ ) [2] and TZNF60 glass ( $\sigma_{\text{ab}} = 0.68$ ,  $\sigma_{\text{em}} = 0.72$ ) [26].

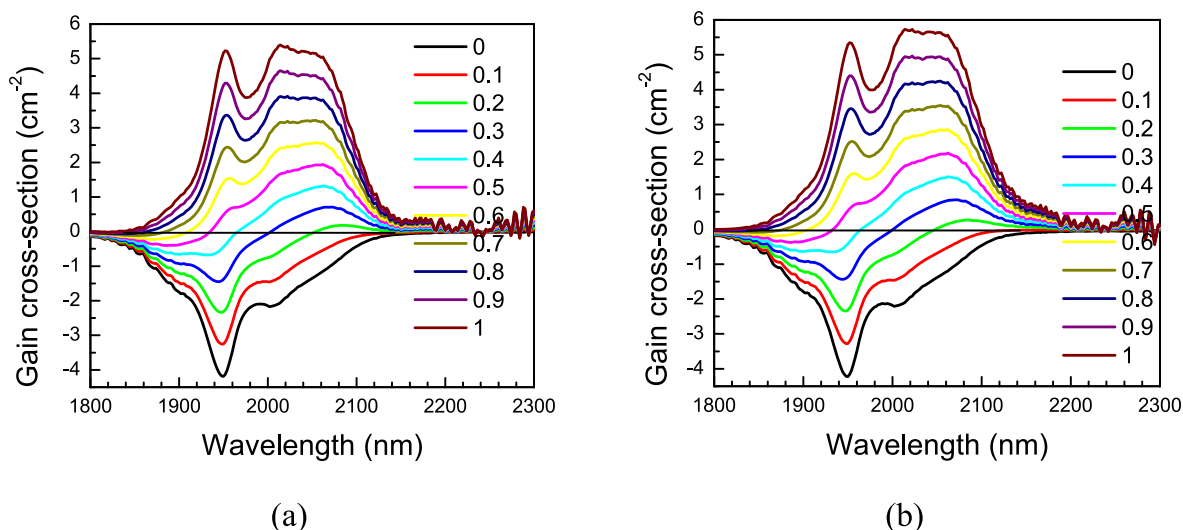
The wavelength dependent gain cross-section can be expressed as:

$$\sigma_g(\lambda) = \gamma \sigma_{\text{em}}(\lambda) - (1 - \gamma) \sigma_{\text{ab}}(\lambda) \quad (2)$$

where  $\sigma_{\text{em}}$  and  $\sigma_{\text{ab}}$  represents emission and absorption cross-sections, respectively, and  $\gamma$  (0–1 with an increments of 0.1) represents the population of  $\text{Ho}^{3+}$  ions. Gain cross-section spectra of the  $^5\text{I}_8 \leftrightarrow ^5\text{I}_7$  transitions of  $\text{Ho}^{3+}$  ion at 2.0  $\mu\text{m}$  as a function of population inversion for a glass and the glass-ceramic GC4 are shown in Fig. 10 (a) and (b), respectively. Gain is positive when the population inversion is equal or greater than 0.3. However, broadening is more predominant for GC4 compared to glass. The gain is found to be  $5.79 \text{ cm}^{-2}$  for GC4 which is higher than that of glass,  $5.35 \text{ cm}^{-2}$ . The value of gain is very high compared to those of reported tellurite [36] and fluorophosphate [37] glasses. It is observed from the gain spectra that a tunable wavelength range of 1900–2100 nm is expected when the population inversion is greater than or equal to 0.5. These results reveal that the  $\text{Ho}^{3+}$ -doped glass-ceramics have potential for optical amplifiers.

## 4. Conclusions

The  $\text{Ho}^{3+}$ - $\text{Yb}^{3+}$  co-doped transparent oxyfluoride glass and glass-ceramics containing  $\text{CaF}_2$  nanocrystals have been investigated. X-ray diffraction patterns confirm the formation of  $\text{CaF}_2$  nanocrystals in the glass matrix. The size of the nanocrystals increases from 8 to 18 nm with increasing the thermal treatment temperature of the precursor glass. Near-infrared quantum cutting as well as 2.0  $\mu\text{m}$  eye safe emissions are obtained in  $\text{Yb}^{3+}$  ions and  $\text{Ho}^{3+}$  ions, respectively by exciting with 473 nm laser at the  $^5\text{F}_2$  level of  $\text{Ho}^{3+}$  ions. The similar emission is also achieved at different delay times by exciting the  $^5\text{F}_4$  level of  $\text{Ho}^{3+}$  ions at 532 nm. The decay curves of  $\text{Yb}^{3+}$  emission evidences the energy transfer process from  $\text{Ho}^{3+}$  to  $\text{Yb}^{3+}$  ions. It is worth noting that lifetime of 1.2  $\mu\text{m}$  emission has been enhanced by three order of magnitude in GC4 sample compared to that of the precursor glass. Intense  $\text{Yb}^{3+}$  and  $\text{Ho}^{3+}$  near-infrared emissions indicate that the present oxyfluoride glass-ceramics containing  $\text{CaF}_2$  nanocrystals have potential application in



**Fig. 10.** Gain cross-section spectra of  $\text{Ho}^{3+}$  ion at 1950 nm in  $\text{Ho}^{3+}/\text{Yb}^{3+}$ -doped oxyfluoride (a) glass and (b) glass-ceramic (GC4).

realizing the solar spectrum to enhance the efficiency of c-Si solar cells.

### Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

### CRediT authorship contribution statement

**P. Babu:** Conceptualization, Investigation, Writing - original draft, Writing - review & editing. **I.R. Martín:** Methodology, Investigation. **V. Lavín:** Investigation, Writing - review & editing. **U.R. Rodríguez-Mendoza:** Formal analysis. **Hyo Jin Seo:** Methodology, Resources. **K. Venkata Krishnaiah:** Software, Data curation. **V. Venkatramu:** Validation.

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### Appendix A. Supplementary data

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